

Does a Discovered Reinforcement Learning Rule Minimise Anything?

A gradient-field audit of DiscoRL

Tianchi Chen

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Abstract

DiscoRL replaces a hand-written reinforcement learning update with a meta-learned one, and the meta-parameters of its strongest rule, Disco103, are public. That makes a question askable that could not be asked of a rule nobody had written down: *is the discovered update the gradient of anything?* The question is not rhetorical. By the Poincaré lemma a vector field is a gradient exactly when its Jacobian is symmetric, so “which objective does this rule minimise” is well posed only if a single scalar, $\kappa = \|\text{sym } J\|_F / \|J\|_F$, equals one.

We measure it. Working in prediction space rather than parameter space, the non-conservative part of the field is isolated in the bootstrap operator $B = \partial \hat{p} / \partial u$ — precisely the term `stop_gradient` discards — and we estimate $\kappa(B)$ by paired Hutchinson probes through a single forward map. Two things make the number readable. First, an analytic reference: a *causal* bootstrap, whose target reads only the future, is strictly triangular, so $\text{tr}(B^2) = 0$ and $\kappa = 1/\sqrt{2}$ exactly, in any dimension. Second, an estimator validated before it is believed, on synthetic fields spanning $\kappa \in [0.025, 1]$ and again in the narrow band where the answer lands, and returning exactly 1.0000 for a genuine gradient.

Across six configurations — two environments, one of them from Disco103’s own meta-training family, and training lengths spanning a factor of sixteen — $\kappa = 0.7084$ to 0.7094 , against a floor of 0.70711 . Disco103’s update is therefore not the gradient of any functional, and it carries no more circulation than causal bootstrapping forces. Three probes of what the discovered predictions represent all return nothing, against a positive control that works. We report the traps that silently return the wrong answer, and two claims of our own that we withdrew.

1 Introduction

A reinforcement learning algorithm is usually an equation. DiscoRL [1] replaces the equation with a network: a meta-network reads a trajectory and emits targets, the agent regresses onto them, and the meta-network itself is trained by meta-gradient across a population of agents. The resulting rule, Disco103, matches or beats hand-designed algorithms on Atari, ProcGen and DMLab, and generalises to environments never seen during discovery. Its meta-parameters ship under an open licence.

That artefact makes a structural question concrete. Hand-designed rules come with a story about what they optimise — policy gradient ascends expected return, trajectory balance [5] descends an explicit residual. A discovered rule comes with no such story. It is natural to ask which objective it implicitly minimises, and a good deal of theory would follow if one could name it.

The question presupposes that an objective exists. It need not. Temporal difference learning famously is not the gradient of any function: the target is held fixed when the gradient is taken, and the discarded term is what a genuine gradient would have included [2, 3]. DiscoRL is built the same way — produce a target, regress onto it — so the presupposition has to be tested rather than assumed.

By the Poincaré lemma the test reduces to one scalar. On a simply connected domain a smooth field v is $-\nabla\Phi$ for some Φ if and only if its Jacobian is symmetric everywhere. Writing $\text{sym } M = (M + M^\top)/2$ and $\text{asym } M = (M - M^\top)/2$, and noting that these are orthogonal in the Frobenius inner product, define

$$\kappa(M) = \frac{\|\text{sym } M\|_F}{\|M\|_F}, \quad \kappa = 1 \iff M \text{ symmetric.} \quad (1)$$

This paper measures κ for Disco103.

Contributions.

1. An analytic reference scale. Any bootstrap whose target reads only the future is strictly triangular, hence has $\kappa = 1/\sqrt{2}$ exactly, independent of dimension and of every other detail (Proposition 1). Without it a measured κ is uninterpretable.
2. A measurement procedure that survives three traps, each of which returns a confidently wrong answer rather than an error (Section 3), with an estimator validated twice before use.
3. The measurement: $\kappa = 0.7084$ – 0.7094 over six configurations, including the environment family Disco103 was meta-trained on. The update is not a gradient field, and sits on the causal floor.
4. Three negative probes of what the discovered predictions carry, with a positive control that works, bounding how strongly the negatives may be read.
5. A withdrawal section. Two interpretations we published in an earlier draft of this work were wrong, and we say which and why.

2 The field and where its asymmetry lives

DiscoRL’s per-step agent loss is a sum of categorical Kullback–Leibler terms between meta-network targets and agent predictions,

$$\mathcal{L}_\theta = c_\pi D_{\text{KL}}(\hat{\pi} \parallel \pi_\theta) + c_y D_{\text{KL}}(\hat{y} \parallel y_\theta) + c_z D_{\text{KL}}(\hat{z} \parallel z_\theta(\cdot, a)) + \cdots \quad (2)$$

with y and z carrying no predefined semantics. In the released implementation $\hat{\pi}, \hat{y}, \hat{z}$ are wrapped in a stop-gradient when the agent update is taken, so the update is a *semi*-gradient.

We work in prediction space rather than parameter space. This is legitimate and also convenient. If a field is a gradient in prediction space it pulls back to a gradient in parameter space, since $-(\partial u / \partial \theta)^\top \nabla_u \Phi = -\nabla_\theta \Phi(u(\theta))$, so the same question is being asked; and prediction space needs only *first* derivatives of the target map, which matters for reasons given in Section 3.

Write u for the prediction logits and $p = \text{softmax}(u)$. The gradient of a categorical KL with respect to the prediction logits, target held fixed, is the familiar difference of probabilities, so the

field is

$$v(u) = \hat{p}(u) - p(u), \quad J = \underbrace{\frac{\partial \hat{p}}{\partial u}}_{=:B} - \underbrace{\frac{\partial p}{\partial u}}_{=\text{diag}(p)-pp^\top}. \quad (3)$$

The second term is symmetric, always. Every departure from a gradient field therefore lives in B , the dependence of the target on the prediction — which is exactly the term the stop-gradient discards. The measurement is $\kappa(B)$.

2.1 An analytic floor

Proposition 1. *For any square M , $\|M \pm M^\top\|_F^2 = 2\|M\|_F^2 \pm 2\text{tr}(M^2)$, hence*

$$\kappa(M)^2 = \frac{1}{2} + \frac{\text{tr}(M^2)}{2\|M\|_F^2}. \quad (4)$$

If M is strictly triangular in some ordering then M^2 is strictly triangular, so $\text{tr}(M^2) = 0$ and $\kappa(M) = 1/\sqrt{2}$ exactly.

Proof. $\langle M, M^\top \rangle_F = \text{tr}(M^\top M^\top) = \text{tr}((M^2)^\top) = \text{tr}(M^2)$, and expanding $\|M \pm M^\top\|_F^2$ gives the identity. Equation (4) follows from $\kappa^2 = \|M + M^\top\|^2 / (\|M + M^\top\|^2 + \|M - M^\top\|^2)$. Strict triangularity gives a nilpotent M with zero diagonal, so $\text{tr}(M^2) = \sum_{ij} M_{ij}M_{ji} = 0$. $\square$

Remark. A *causal* bootstrap — one whose target at index t depends only on indices after t — is strictly triangular in the time ordering. So $\kappa = 1/\sqrt{2} \approx 0.70711$ is the structural floor imposed by the arrow of time in credit assignment, and it is dimension-free. Tabular TD is the canonical instance: its target map is $\hat{V} = r + \gamma PV$, so $B = \gamma P$ and on any acyclic graph $\kappa = 1/\sqrt{2}$.

The floor is not forced for DiscoRL. Its meta-network receives its own y at index t as an input — the released input specification pairs y_t with y_{t+1} — so B has a nonzero diagonal block and $\text{tr}(B^2) \neq 0$ in general. Whether Disco103 nonetheless sits on the floor is an empirical question, and by (4) the excess over the floor is measurable directly as $\text{tr}(B^2)/\|B\|_F^2 = 2\kappa^2 - 1$.

2.2 Estimator

Frobenius norms come from paired Hutchinson probes: for $w \sim \mathcal{N}(0, I)$, $\|M\|_F^2 = \mathbb{E}_w \|Mw\|^2$, with Bw from a Jacobian-vector product and $B^\top w$ from a vector-Jacobian product *on the same w* . Pairing matters: the two norms are strongly correlated, so their ratio is far better conditioned than either term. Standard errors come from a block jackknife over probes.

Validation, twice, before use. On synthetic fields $g(x) = (S + cA)x$ with S symmetric and A antisymmetric, κ is known in closed form; sweeping c walks the estimator across $\kappa \in [0.025, 1]$ with worst absolute error 0.0036, and it returns *exactly* 1.0000 at $c = 0$. Since the answer lands near $1/\sqrt{2}$, we calibrate again there: strictly triangular B_0 plus a controlled diagonal block, where (4) gives the exact value, with worst error 0.0041. That 0.0041 is the resolution with which any excess over the floor can be claimed, and we quote it whenever we compare against the floor.

3 Three traps

Each of the following returns a plausible number rather than an error, and each was hit before it was fixed.

Environment	Training steps	$\kappa(B)$	$\text{tr}(B^2)/\ B\ _F^2$	$\kappa - 1/\sqrt{2}$
hypergrid	100	0.7084 ± 0.0002	+0.00358	+0.0013
hypergrid	400	0.7086 ± 0.0007	+0.00435	+0.0015
hypergrid	1600	0.7090 ± 0.0009	+0.00550	+0.0019
Catch	100	0.7086 ± 0.0003	+0.00414	+0.0015
Catch	400	0.7088 ± 0.0004	+0.00477	+0.0017
Catch	1600	0.7094 ± 0.0008	+0.00648	+0.0023
causal floor $1/\sqrt{2}$		0.70711	0	—
estimator resolution in this band		0.0041		

Table 1: κ of Disco103’s bootstrap operator. Dimension 86,616, 48 paired probes, errors by block jackknife. Every excess over the floor is smaller than the estimator’s demonstrated resolution in the same band.

Second-order stop-gradient. `jax.lax.stop_gradient` has zero derivative at *all* orders. Differentiating the released `backprop=False` path a second time therefore erases B — the very object under measurement — and reports a perfect gradient field. Our first attempt returned $\beta = 0.0000$ from an autodiff path for exactly this reason. The cure here is structural: B is obtained as a *first* derivative of the target map, so no stop-gradient is ever on the path.

Querying an untrained agent. Disco103 was meta-learned against competent agents. Presented with a freshly initialised one — near-uniform policy, near-uniform 600-way predictions — its outputs collapse towards a constant and every sensitivity reads as zero for reasons unrelated to conservativity. The agent must first be trained *by Disco103 itself*.

Silent out-of-bounds. JAX drops out-of-bounds scatter indices silently. A control that perturbs index $t + 3$ while reading index t becomes a no-op at the end of a rollout, and the resulting 0/0 surfaces as `nan` rather than as an error. Relatedly, an ε placed outside rather than inside the square root in an unrolled Adam gives an infinite derivative at zero second moment — which masked-out logits always have — turning the outer gradient into `nan` while the forward pass stays finite.

4 Result

We train an agent with Disco103 on a given environment, freeze a rollout, and measure $\kappa(B)$ on the joint prediction vector of policy logits, y logits, and z logits at the action taken; the dimension is 86,616. Two environments are used. The first is a hypergrid, a DAG in which several action sequences reach the same state. The second is Catch, which belongs to the family Disco103 was actually meta-trained against and therefore answers the obvious objection that the first is an out-of-distribution query. Training length is swept over a factor of sixteen.

Table 1 gives the result. Two readings, both of which matter.

$\kappa \neq 1$: the update is not a gradient field. The estimator returns exactly 1.0000 for a genuine gradient, and the frozen-target control on the same network — the same loss with the target held constant, which is an honest scalar objective — also returns 1.0000. Against that, 0.7088 is not a

Readout	Target	R^2	Spearman	Role
q head	V_π	0.559	+0.844	positive control
$\varphi(z)$	V_π	0.478	−0.722	
$\varphi(y)$	V_π	0.125	−0.377	
$\varphi(y)$	$\log F$	0.018	−0.080	flow hypothesis
$\varphi(y)$	$\mathcal{H}_{\text{future}}$	0.002	—	entropy hypothesis

Table 2: State-conditional means over 64/64 visited states, 40 rollouts, exact targets by dynamic programming. The control works; the y channel tracks nothing.

near miss. Disco103’s update is not $-\nabla\Phi$ for any Φ , so *which objective does DiscoRL minimise* is malformed rather than open.

$\kappa \approx 1/\sqrt{2}$: **no more circulation than causality forces.** Every excess over the floor lies between +0.0013 and +0.0023, below the 0.0041 resolution demonstrated in that band. The non-conservativity is therefore not something meta-learning discovered; it is the generic price of a causal bootstrap. The excess is nonzero and increases monotonically with training length in both environments, consistent with the diagonal block that the paired (y_t, y_{t+1}) input creates, but it remains within resolution throughout.

What can be said about the conservative part. By orthogonality of sym and asym, the Hodge projection onto the conservative sector retains $\kappa^2 \approx 0.502$ of the operator’s Frobenius mass. We state this at the level of the *operator* only. Connecting an operator-mass split to sample efficiency or solution quality is work we have not done, and the number should not be read as “half of DiscoRL is a gradient method”.

5 What do the discovered predictions represent?

DiscoRL’s own analysis reports that the learned predictions track future policy entropy and upcoming large-reward events. On a hypergrid the natural candidate state functions are computable exactly by dynamic programming, so the claim can be tested rather than paraphrased. We use Disco103’s *own* scalar readout of its 600-way prediction — the `y_net` and `z_net` multilayer perceptrons whose weights ship with the released parameters — so the probe has zero free parameters. The categorical q head, a value head by construction, serves as positive control.

Table 2 reports the outcome. The control passes cleanly, so the bucketing, the training and the dynamic programming all work; against that baseline the y channel tracks neither a flow, nor a value, nor future policy entropy. A joint regression does not rescue it: adding $\varphi(y)$ to $\varphi(z)$ buys 0.010 of R^2 on $\log F$ but 0.060 on the pure value control, so it behaves as a generic weak regressor rather than as the complementary half of a free-energy decomposition.

How strongly may these nulls be read? A potential must be a state function, and the agent’s torso is recurrent, so we decompose each readout’s variance. $\varphi(y)$ is a *pure* state function, $\eta^2 = 1.0000$ — the bucketing was never washing it out — but its dynamic range is $68\times$ smaller than $\varphi(z)$ ’s. The honest scope is therefore: in this regime the y channel carries a deterministic but vanishingly small state signal that matches no candidate potential. That is weaker than “ y can never be a flow”.

A sensitivity that is real. A separate probe shows the y -target does respond to the log-probability of the action taken, at $139\times$ the random-initialisation null, and the response is local: moving the perturbation three steps later reduces it by $23\times$. The y channel is not inert as a map; it is unidentified as a function.

6 Withdrawals

An earlier draft of this work reported two interpretations that we now retract. Both arose from the same failure: reading a number that has a simple structural explanation as evidence for a more elaborate one.

First, we reported the ratio of the y -target’s sensitivity to log-policy against its sensitivity to the successor prediction, $|\beta/\alpha| = 0.32$, as “a third of the way to detailed balance”. That reading presumes $\varphi(y)$ is a log-flow quantity. The calibration in Table 2 refutes the premise, so the reading is withdrawn; the sensitivity measurement itself stands.

Second, we read $\kappa^2 \approx 0.5$ as evidence that the update implements a saddle-point, primal–dual scheme, on the grounds that such schemes have an irreducibly antisymmetric coupling block. But Proposition 1 already accounts for the same number with no saddle structure whatsoever, and it was derived first. The simpler explanation suffices; the saddle reading is withdrawn.

7 Related work, and what is not ours

The gradient-field question for bootstrapped updates is old. Residual-gradient methods differentiate through the target and are genuine gradient methods, at the cost of a double-sampling bias in stochastic environments and a different, often worse, fixed point under function approximation [2]; the gradient-TD family restores a true objective by introducing an auxiliary variable [3], and the convex-duality view organises this [4]. What is new here is not the question but its application to a rule that was discovered rather than designed, together with an analytic reference scale that makes the answer readable.

We also record what is *not* ours. In the course of this work we derived a structural result about GFlowNet flow spaces — that the set of R -flows is an affine set directed by the cycle space of the state graph, so the backward policy is a gauge freedom that cannot move the terminal distribution. It is published. Brunswic et al. [6] prove $\mathcal{F}_R = F + H^1(G)$ and attribute a stronger form to Kalpazidou; that every backward policy induces the same terminal distribution is standard and stated in at least six papers, most crisply in trajectory balance [5], which already names cycles in the underlying undirected graph as the source of the multiplicity; and learning the backward policy for performance with correctness preserved by construction has been done with at least five distinct objectives. Our addition was the dimension $|E| - |V| + 1$ and its identification with the parameter count of the backward policy, which is the first Betti number and follows immediately. We claim none of it.

We note also that the method we had been using to judge novelty — counting arXiv abstract matches — failed on exactly this case, because the pre-empting paper writes $H^1(G)$ in cohomological notation and never uses the phrase “cycle space”.

8 Limitations

The measurement is of one rule, Disco103, on two environments. Catch answers the distribution objection but neither environment is ProcGen or DMLab. κ is computed on a frozen rollout at a single point in parameter space, and while it is stable across a sixteen-fold range of training length, it is a local quantity. The probe restricts to the prediction channels the agent loss touches and holds the Polyak-delayed target network fixed, which is faithful within one update but suppresses one dependency. Batch elements couple at $O(1/B)$ through the advantage and TD normalisers. Finally, the conservative-mass statement $\kappa^2 \approx 0.5$ is an operator-level fact with no demonstrated behavioural consequence.

9 Reproducibility

Every number in this paper is produced by a script, and the scripts are in the accompanying repository. `kappa.py` performs the estimator validation and the measurement; `calibrate.py` and `thermo.py` perform the prediction probes and the variance decomposition; `disco_probe.py` performs the sensitivity measurement; `verify_disco_source.py` encodes the claims we make about the released implementation as executable assertions pinned to commit 9059a29f, re-run weekly, so that a claim about someone else’s code cannot rot silently. All exact quantities are computed by dynamic programming over the state graph rather than sampled.

References

- [1] J. Oh, G. Farquhar, I. Kemaev, D. A. Calian, M. Hessel, L. Zintgraf, S. Singh, H. van Hasselt, D. Silver. Discovering state-of-the-art reinforcement learning algorithms. *Nature*, 2025. <https://doi.org/10.1038/s41586-025-09761-x>
- [2] L. Baird. Residual algorithms: reinforcement learning with function approximation. *ICML*, 1995.
- [3] R. S. Sutton, H. R. Maei, D. Precup, S. Bhatnagar, D. Silver, C. Szepesvári, E. Wiewiora. Fast gradient-descent methods for temporal-difference learning with linear function approximation. *ICML*, 2009.
- [4] O. Nachum, B. Dai. Reinforcement learning via Fenchel–Rockafellar duality. arXiv:2001.01866, 2020.
- [5] N. Malkin, M. Jain, E. Bengio, C. Sun, Y. Bengio. Trajectory balance: improved credit assignment in GFlowNets. *NeurIPS*, 2022. arXiv:2201.13259.
- [6] L. Brunswic, Y. Li, Y. Xu, S. Jui, L. Ma. A theory of non-acyclic generative flow networks. *AAAI*, 2024. arXiv:2312.15246.
- [7] D. Tiapkin, N. Morozov, A. Naumov, D. Vetrov. Generative flow networks as entropy-regularized RL. *AISTATS*, 2024. arXiv:2310.12934.